

Samir Darouich

PhD researcher in machine learning for chemistry, University of Stuttgart

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[Website](#) | [Google Scholar](#) | [GitHub](#) | [LinkedIn](#)

Research interests

Generative models (flow matching, diffusion) for chemistry; transition states and chemical reactivity; symmetry-aware and physics-informed machine learning; molecular simulation.

Education

PhD in Theoretical Chemistry and Machine Learning

06/2024 – present

University of Stuttgart

Stuttgart, Germany

Thesis: *Machine Learning for Reactivity Elucidation in Computational Chemistry*. Advisors: Prof. J. Kästner (Institute for Theoretical Chemistry) and Prof. M. Niepert (Institute for Artificial Intelligence).

MSc in Chemical and Biological Engineering (grade 1.0)

10/2021 – 09/2023

University of Stuttgart

Stuttgart, Germany

Thesis: *Vapor–liquid equilibria of binary mixtures estimated from molecular dynamics free energy simulations in the liquid phase*. Advisors: Prof. N. Hansen and Prof. J. Gross.

BSc in Chemical and Biological Engineering (grade 1.7)

10/2018 – 09/2021

University of Stuttgart

Stuttgart, Germany

Thesis: *Transferability of the TAMie force field for primary alcohols to secondary alcohols and diols*. Advisors: Prof. N. Hansen and Prof. J. Gross.

Research experience

Doctoral researcher

06/2024 – present

University of Stuttgart, Institutes for Theoretical Chemistry and for AI

Stuttgart, Germany

- Introduced adaptive equilibrium flow matching (AEFM), which refines low-fidelity TS guesses, increasing successful localization by 41% and speeding up high-level quantum optimization threefold (*JCIM* 2026).
- Developed SymDrift, a symmetry-aware one-shot generative model for molecules that is competitive with multi-step baselines on conformer and TS generation while being up to 40× faster (*NeurIPS* 2026).
- Proposed CurvFM, a flow matching objective that constrains the curvature of the energy surface at generated structures, reducing the median maximum-force deviation by 63% on GEOM-QM9 and raising the share of valid TSs by up to 10% (preprint 2026).

Visiting PhD student, Kulik group

09/2025 – 11/2025

Department of Chemical Engineering, Massachusetts Institute of Technology

Cambridge, USA

- Designed benchmarks with unseen elements and transition metal complexes that exposed systematic failure modes of generative TS models and developed a self-supervised pretraining strategy on equilibrium conformers that improves accuracy on unseen chemistry while cutting the required fine-tuning data by up to 75% (*Nature Computational Science* 2026).
- Contributed to two further collaborations, on ElemeNet, a multiscale molecular machine learning package with uncertainty quantification across the periodic table (*JCIM*, accepted), and on physics-based molecular fingerprints from spectral graph theory for geometry-aware chemical similarity (preprint 2026).

Data steward, Collaborative Research Center 1333

10/2023 – 05/2024

University of Stuttgart

Stuttgart, Germany

- Established FAIR data management in chemistry by developing data models and workflows to acquire, analyze and publish data on Dataverse.
- Developed a Python module to set up and analyze molecular dynamics simulations.

Intern and working student, Molecular Modeling & Drug Discovery

BASF SE

10/2022 – 09/2023

Ludwigshafen, Germany

- Devised a method to scan adsorption free energies of organic molecules on inorganic mineral surfaces with all-atom force fields that BASF adopted into its modeling workflows, and trained BASF modeling researchers in its use (*Corros. Sci.* 2024).
- Simulated the adsorption of organic additives on metal oxides and cement minerals, guiding the selection of the most effective additives (*Cem. Concr. Res.* 2025).
- Studied the interfacial tension between tricalcium aluminate and water, which led to a prediction of the best salt additives for inhibiting its reaction with water (*J. Am. Ceram. Soc.* 2026).
- Built tools for all-atom force field development and validation and applied them to create an INTERFACE force field for alumina with validated bulk phases and a pH-resolved surface model database that enables simulations of electrolyte and organic interfaces (*Chem. Eng. J.*, accepted).

Pro3 research fellow

Institute of Technical Thermodynamics, University of Stuttgart

04/2022 – 10/2022

Stuttgart, Germany

- Analyzed conformers of diols with quantum mechanical methods to develop new TAMie force field parameters (*J. Phys. Chem. B* 2024).
- Applied machine learning to force field parameterization to handle sparse experimental data (*J. Phys. Chem. B* 2024).

Selected publications

1. **Darouich, S.**, Tong, V., Pastor-Pérez, L., Bien, T., Mualem, L., & Niepert, M. (2026). SymDrift: One-shot generative modeling under symmetries. *NeurIPS*
2. **Darouich, S.**, Toney, J. W., Luo, W., Kästner, J., Niepert, M., & Kulik, H. J. (2026). Robust generative transition-state models for unseen chemistry. *Nat. Comput. Sci.*
3. **Darouich, S.**, Tong, V., Bien, T., Kästner, J., & Niepert, M. (2026). Adaptive transition-state refinement with learned equilibrium flows. *J. Chem. Inf. Model.*, 66(4), 2154–2165.
4. **Darouich, S.**, Geisler, J., Toney, J. W., Bien, T., Kästner, J., Kulik, H. J., & Niepert, M. (2026). Curvature-aware flow matching for molecular structure generation. *arXiv*, arXiv:2609.34592.
5. Toney, J. W., **Darouich, S.**, Wang, Y., Garrison, A. G., Kästner, J., & Kulik, H. J. (2026). ElemeNet: Multiscale molecular machine learning with uncertainty quantification across the periodic table. *J. Chem. Inf. Model.*
6. Toney, J. W., Farnood, A. Y., **Darouich, S.**, & Kulik, H. J. (2026). Physics-based molecular fingerprints from spectral graph theory provide efficient geometry-aware measures of chemical similarity. *arXiv*, arxiv:2608.05336
7. Mualem, L., Tong, V., **Darouich, S.**, & Niepert, M. (2026). ARIA: Adaptive region-based importance allocation for conditional diffusion distillation. *arXiv*, [arxiv:2606.23898](https://arxiv.org/abs/2606.23898)
8. Fleck, M., Spera, M. B. M., **Darouich, S.**, Klenk, T., & Hansen, N. (2025). Generalized DeepONets for viscosity prediction using learned entropy scaling references. *Digit. Discov.*, 4(12), 3578–3587.
9. Fleck, M., **Darouich, S.**, Pleiss, J., Hansen, N., & Spera, M. B. M. (2025). Physics-informed multifidelity Gaussian process: Modeling the effect of water and temperature on the viscosity of a deep eutectic solvent. *J. Chem. Inf. Model.*, 65(8), 3999–4009.

Conference contributions

Computational Chemistry, Gordon Research Conference, Barcelona, Spain

2026

Poster: Robust generative transition-state models for unseen chemistry.

Molecular Machine Learning Conference (MoML), MIT, Cambridge, USA

2025

Poster: Adaptive transition-state refinement with learned equilibrium flows.

AI4Science Workshop, NeurIPS, San Diego, USA

2025

Poster: Adaptive transition-state refinement with learned equilibrium flows.

Teaching and outreach

Seminar instructor, Machine Learning for Science

ongoing

Institute for Artificial Intelligence, University of Stuttgart

Stuttgart, Germany

- Guide students in reading a research paper and preparing a presentation and blog post on it.

Teaching assistant, Advanced Theoretical Chemistry II

12/2024 – 02/2025

Institute for Theoretical Chemistry, University of Stuttgart

Stuttgart, Germany

- Taught weekly tutorials in advanced theoretical chemistry.

Teaching assistant, Technical Thermodynamics

10/2020 – 03/2022

Institute of Technical Thermodynamics, University of Stuttgart

Stuttgart, Germany

- Taught weekly tutorials in technical thermodynamics for a group of 20 undergraduate students, including grading.
- Mentored two groups of 25 undergraduate students each in advanced engineering mathematics, reviewing homework and exams.

BOGY career orientation program

2025

University of Stuttgart

Stuttgart, Germany

Designed and ran a program introducing high school students to theoretical chemistry.

Service

Organizing team, research group retreat

2026

Institute for Artificial Intelligence, University of Stuttgart

Hirschegg, Austria

Organizing team, AISA Symposium

2025

Artificial Intelligence Software Academy, University of Stuttgart

Stuttgart, Germany

Honors and awards

2022 Pro3 Research Fellowship, University of Stuttgart

2021 Annual Program of the School for Talents, University of Stuttgart

2021 Teaching assistant qualification

Additional professional experience

Working student, Process Technology

09/2021 – 03/2022

Dürr Systems AG

Bietigheim-Bissingen, Germany

- Worked on CO₂ separation from exhaust gas using bipolar membrane technology and built a prototype for measuring gas–liquid CO₂ mass transfer.

Working student, Exhaust Aftertreatment Systems

11/2019 – 10/2020

Robert Bosch GmbH

Stuttgart, Germany

- Prepared diesel exhaust catalyst samples and developed test procedures to characterize greenhouse gas emission reduction.

Skills

Machine learning Flow matching, diffusion models, geometric and equivariant deep learning, Gaussian process regression, multi-fidelity modeling

Computational chemistry DFT, molecular dynamics, Monte Carlo simulations, force field development

Software Python, PyTorch, C/C++, Git/GitHub, DVC, LAMMPS, GROMACS, HPC clusters

Languages German (native), English (C1), French (B1), Spanish (A2)

References

Prof. Dr. Johannes Kästner Institute for Theoretical Chemistry, University of Stuttgart

Prof. Dr. Mathias Niepert Institute for Artificial Intelligence, University of Stuttgart

Prof. Heather J. Kulik Department of Chemical Engineering, MIT